

A Packing of 19 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 19 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.532950. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.525899, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 19 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.532950 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 19$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.525899, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 19/18 = 2.533333$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N19.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
140719475011004759789193872148738936373343044936448863358147830958140157
558904058737314603250402451465430735099236062730602106755063628002636494
458960638850824139091646573474979793847396335551
```

denominator:

```
55555555555023532286387597615025638044493774326002953980805641408183145
608756040685258786702985248397214654316007005884408462469125664812643154
97119157303886907938452172073909714547623509000
```

B Coordinates

19 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N19.txt`.

```
-266702223923 134590018710 379949664892 -284704306363 2264373002 16433450832 20620653051 -23580556261 292565653964 -177184619155 149360111361 82750525952
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